Gradient boosting for regression

Squared-error loss, regression trees as base learners

Data: eight points

Summary.
1. Start with a very rough prediction: the same number for everybody (the average).
2. Look at how wrong that prediction is for each data point. These errors are the **residuals**.
3. Learn a tiny model (a *stump*: one yes/no question) that predicts the residuals, and add a *small fraction* of it to the prediction.
4. Look at the new errors and repeat. The final prediction is the starting number plus all the small corrections added up.

Colour key: every symbol keeps its colour in every formula

y	**the truth**	the observed value we want to predict
F	**the prediction**	the model's current guess; starts at the average
r	**the residual**	$r = y - F$: how far the truth is above the prediction
γ	**the leaf value**	the correction a leaf adds: the average residual of its points
ν	**the step size**	the fraction of the correction we actually add (here 0.5)
g	**the slope**	slope of the loss with respect to F (used only in the boxes)
h	**the curvature**	second derivative of the loss (used only in the boxes)

1. The problem

Regression means predicting a *number* (not a class). Here is a small, *made-up, illustrative* data set: how long one training epoch takes (in minutes) for a neural network of a given size.

point i	model size x (millions of parameters)	time per epoch y (minutes)
1	1	2
2	2	3
3	3	3
4	4	5
5	5	6
6	6	8
7	7	8
8	8	9

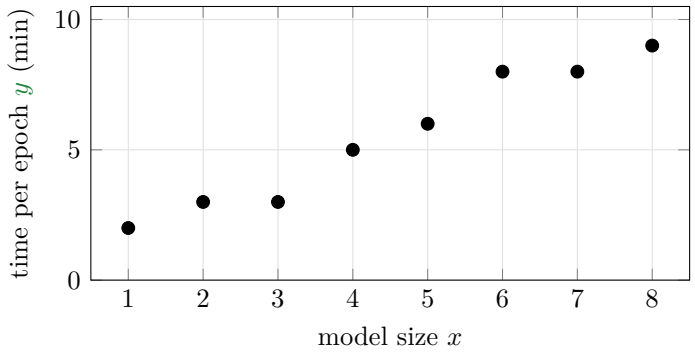


Figure 1. The eight points. Bigger models take longer, but not along a straight line.

We want a function $F(x)$ that predicts y from x. Three words we use all the time:

- **Prediction** $F(x_i)$: what the model says for point i.
- **Residual** $r_i = y_i - F(x_i)$: how far the truth is above the prediction. Positive: we predicted too low. Negative: too high.
- **Loss**: one number measuring how wrong the model is. We use the squared error $\frac{1}{2}\sum_i (y_i - F(x_i))^2$: wrong by 2 costs four times as much as wrong by 1. Big mistakes are punished.

2. Step 0: the first guess is the average

Before any tree, the model gives *every* point the same prediction F_0. Which single number is best? The one that makes the squared error smallest. The answer is the **average** of the y's:

$$F_0 = \frac{2+3+3+5+6+8+8+9}{8} = \frac{44}{8} = 5.5.$$

Box 1: why the average?

Let the same prediction c be used for every point. Write the squared error $S(c) = \frac{1}{2}\sum_{i=1}^{N}(y_i - c)^2$, differentiate with respect to c, set it to zero and solve.

Hint: the derivative of $\frac{1}{2}(y_i - c)^2$ with respect to c is $-(y_i - c)$; then sum over i.

i	x	y	prediction F_0	residual $r = y - F_0$	r^2
1	1	2	5.5	−3.50	12.25
2	2	3	5.5	−2.50	6.25
3	3	3	5.5	−2.50	6.25
4	4	5	5.5	−0.50	0.25
5	5	6	5.5	+0.50	0.25
6	6	8	5.5	+2.50	6.25
7	7	8	5.5	+2.50	6.25
8	8	9	5.5	+3.50	12.25
				sum of squared residuals:	50.00

The residuals are the errors that are left. The small points (low x) are predicted too high (negative residuals), the large ones too low (positive). **The next step tries to predict these residuals.**

3. Round 1

1 Step 1. The helper: a stump. A stump asks *one* question about x, such as "is $x \le 4.5$?", and has two leaves. Each leaf holds one number: the **average residual** of the points that land in it. (The stump is a tiny model that predicts the *errors*, not y.)

2 Step 2. Which question? We try every possible threshold: between each pair of neighbouring x values, so $1.5, 2.5, \dots, 7.5$, seven in all. For each question we compute

$$\text{score} = \frac{\left(\sum_{i\in L} r_i\right)^2}{n_L} + \frac{\left(\sum_{i\in R} r_i\right)^2}{n_R}, \qquad L = \text{points answering yes},\ R = \text{no}.$$

Read it like this: a good question puts the points with positive residuals on one side and the negative ones on the other, so that each side's residuals add up to a big number (not cancel). The question with the **largest score** wins. (Box D4 shows exactly why this score is the right one.)

question	n_L	$\sum_L r$	mean $\bar r_L$	n_R	$\sum_R r$	mean $\bar r_R$	score	
$x \le 1.5$	1	−3.50	−3.500	7	+3.50	+0.500	14.00	
$x \le 2.5$	2	−6.00	−3.000	6	+6.00	+1.000	24.00	
$x \le 3.5$	3	−8.50	−2.833	5	+8.50	+1.700	38.53	
$x \le 4.5$	4	−9.00	−2.250	4	+9.00	+2.250	40.50	**best**
$x \le 5.5$	5	−8.50	−1.700	3	+8.50	+2.833	38.53	
$x \le 6.5$	6	−6.00	−1.000	2	+6.00	+3.000	24.00	
$x \le 7.5$	7	−3.50	−0.500	1	+3.50	+3.500	14.00	

Your turn 1: score one question

Compute the score of the question $x \le 2.5$ yourself from the residuals of Section 2: add the residuals of the points answering yes, add those answering no, then use the formula. Is it larger or smaller than the best score?

Hint: yes: points 1 and 2 (−3.5 and −2.5); no: points 3 to 8. The two sums are opposite because all the residuals add up to 0.

3 Step 3. The leaf values. The winner is $x \le 4.5$. Its leaves hold the average residuals: yes-leaf (points 1–4) $\gamma_L = \dfrac{-9}{4} = -2.25$; no-leaf (points 5–8) $\gamma_R = \dfrac{+9}{4} = 2.25$. In words: the small models are predicted 2.25 minutes too high, the large ones 2.25 minutes too low.

Why the average residual? Write the loss and differentiate. For one data point the loss is $L = \frac{1}{2}(y - F)^2$. Differentiate it with respect to the prediction F (chain rule: the inside $y - F$ has derivative -1):

$$g = \frac{dL}{dF} = (y - F)\cdot(-1) = -(y - F) = -r, \qquad h = \frac{d^2L}{dF^2} = \frac{d}{dF}\big[-(y - F)\big] = +1.$$

So the slope of the loss is $g = -r$ (minus the residual) and the curvature is $h = 1$. Now add the same number γ to the prediction of every point in one leaf. The leaf's total loss, as a function of γ, is

$$T(\gamma) = \sum_{i\in\text{leaf}} \tfrac{1}{2}\,(y_i - F_i - \gamma)^2 = \sum_{i\in\text{leaf}} \tfrac{1}{2}\,(r_i - \gamma)^2.$$

Differentiate with respect to γ, set to zero, and solve:

$$T'(\gamma) = -\sum_{i\in\text{leaf}} (r_i - \gamma) = -\sum r_i + n_{\text{leaf}}\,\gamma = 0 \quad\Longrightarrow\quad \gamma = \frac{\sum_{i\in\text{leaf}} r_i}{n_{\text{leaf}}} = \bar r_{\text{leaf}}.$$

The second derivative is $T''(\gamma) = n_{\text{leaf}} > 0$, so this is a minimum. In the language of slope and curvature this is $\gamma = -\dfrac{\sum g}{\sum h} = \dfrac{\sum r}{n}$. Because the squared error is *exactly* a parabola in γ, no approximation was needed. In the classification notes the loss is not a parabola, and the same recipe (slope over curvature) is applied to a parabola that only approximates it.

4 Step 4. Add a fraction of the correction. We do not add the whole γ. We add $\nu\gamma$ with the **step size** (shrinkage) $\nu = 0.5$:

$$F_1(x) = F_0 + \nu\,\gamma_{\text{leaf}(x)} \quad \Rightarrow \quad \begin{cases} F_1 = 5.5 + 0.5(-2.25) = 4.375 & (x \le 4.5), \\ F_1 = 5.5 + 0.5(2.25) = 6.625 & (x > 4.5). \end{cases}$$

Why only half? One tree is a crude model; trusting it fully makes the next tree start from a worse place if it was fooled by noise. Small steps, many of them, are more reliable.

Your turn 2: update

What is the prediction $F_1(6)$ after round 1 with $\nu = 0.5$? And what would it be with $\nu = 1$ (the full step)? Which one is closer to the truth $y = 8$?

Hint: point 6 is in the no-leaf ($\gamma_R = 2.25$); $F_1 = F_0 + \nu\gamma_R$.



Figure 2. Left: the first guess, one flat line at the average. Right: after round 1 the prediction is a two-step staircase, one step down for small models and one up for large ones.

4. Rounds 2 and 3: the same recipe on the new errors

The recipe never changes: (1) compute the residuals $r = y - F$ of the *current* model, (2) choose the question with the largest score, (3) set the leaf values to the average residual in each leaf, (4) add $\nu\gamma$. After round 1 the residuals are:

i	x	y	F_1	residual $r = y - F_1$
1	1	2	4.375	−2.375
2	2	3	4.375	−1.375
3	3	3	4.375	−1.375
4	4	5	4.375	+0.625
5	5	6	6.625	−0.625
6	6	8	6.625	+1.375
7	7	8	6.625	+1.375
8	8	9	6.625	+2.375

The squared errors have shrunk (the sum of r^2 was 50, now 19.625) but the residuals still show a pattern: points 1–3 are predicted too high, points 4 and 6–8 too low, point 5 slightly too high. The best question is now $x \le 3.5$ (score 14.01), with $\gamma_L = -1.708$ and $\gamma_R = 1.025$.

Your turn 3: the leaf values of round 2

The question of round 2 is $x \le 3.5$. Using the residuals in the table, compute γ_L (the average residual of points 1, 2, 3) and γ_R (the average residual of points 4 to 8). Then the new prediction of point 5: $F_2(5) = F_1(5) + 0.5\,\gamma_R$.

Hint: $\gamma_L = (r_1 + r_2 + r_3)/3$*; the five points of the no-leaf are 4, 5, 6, 7, 8.*

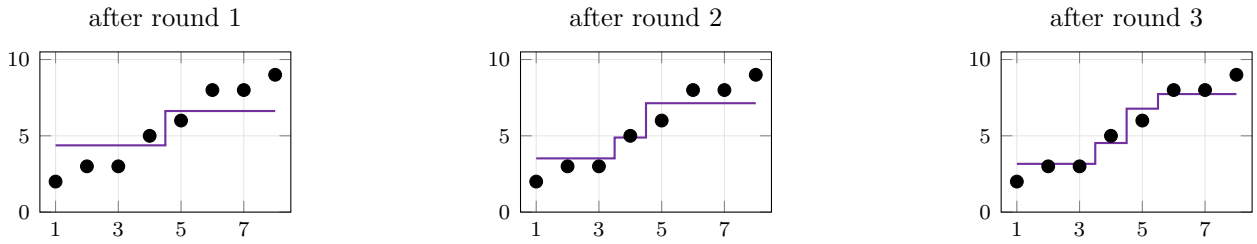


Figure 3. The prediction (red staircase) after rounds 1, 2, 3. Each round adds a small step where the previous staircase was still wrong, so the staircase hugs the points more closely.

round	question	γ_L	γ_R	sum of squared residuals after the round
0	(start)			50.000
1	$x \le 4.5$	−2.250	+2.250	19.625
2	$x \le 3.5$	−1.708	+1.025	9.119
3	$x \le 5.5$	−0.718	+1.196	3.971
4	$x \le 7.5$	−0.181	+1.265	2.600
5	$x \le 1.5$	−1.072	+0.153	1.615
6	$x \le 5.5$	−0.222	+0.371	1.120
7	$x \le 1.5$	−0.425	+0.061	0.966
8	$x \le 7.5$	−0.049	+0.340	0.867

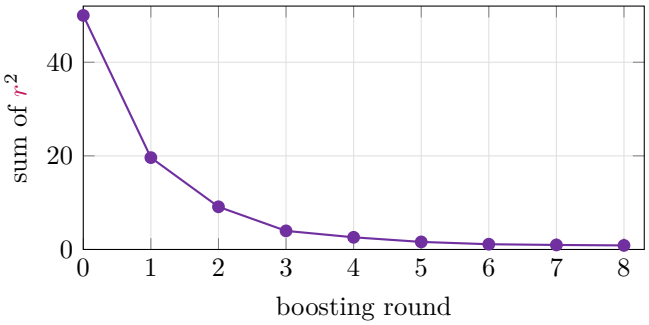


Figure 4. The squared error falls every round, fast at first and then slowly: the big errors are fixed early.

5. The finished model, and predicting a new point

After T rounds the model is the starting number plus the corrections of all the stumps:

$$F(x) = F_0 + \nu \sum_{k=1}^{T} \gamma_k(x)$$

where $\gamma_k(x)$ is the leaf value of stump k that x falls into. For a new model size $x = 2.5$ (not in the data), four rounds:

round	question	$x = 2.5$ answers	adds $\nu\gamma$	running prediction
0	start			5.500
1	$x \le 4.5$	yes	-1.125	4.375
2	$x \le 3.5$	yes	-0.854	3.521
3	$x \le 5.5$	yes	-0.359	3.162
4	$x \le 7.5$	yes	-0.090	3.072

After eight rounds the prediction at $x = 2.5$ is 3.043 minutes, very close to the two neighbouring data points, which both have $y = 3$.

6. Derive it yourself

Each formula above can be derived in two or three lines. Work through the boxes in order; the hints are only nudges and the full working is in Section 9. We write the loss of one point as $L = \frac{1}{2}(y - F)^2$.

Box D1: the residual is the negative slope

Differentiate $L = \frac{1}{2}(y - F)^2$ with respect to the prediction F. What do you get, written with the residual $r = y - F$? What does it mean for the direction we should move F?

Hint: chain rule: $\frac{d}{dF}\frac{1}{2}u^2 = u\,\frac{du}{dF}$ *with* $u = y - F$.

Box D2: the leaf value is the average residual

A leaf contains points $i \in$ leaf and we add the same number γ to all their predictions. The loss of the leaf becomes $\frac{1}{2}\sum_{i\in\text{leaf}}(r_i - \gamma)^2$. Find the γ that minimises it.

Hint: this is Box 1 again, with r_i *instead of* y_i.

Box D3: how much does a stump help?

With $\gamma_L = \bar{r}_L$ and $\gamma_R = \bar{r}_R$ (full step), show that the sum of squared residuals falls by $n_L\bar{r}_L^2 + n_R\bar{r}_R^2$. Check with round 1: the fall should be $50 - 9.5 = 40.5$.

Hint: for one leaf, $\sum(r_i - \bar{r})^2 = \sum r_i^2 - n\bar{r}^2$. *(Then use this for both leaves and compare with no split, using that* $\sum r_i = 0$ *overall.)*

Box D4: why that score picks the best question

The fall in squared error from Box D3, written as $\frac{(\sum_L r)^2}{n_L} + \frac{(\sum_R r)^2}{n_R} - n\bar{r}^2$, has a last term that does *not* depend on the question. So what should we maximise to choose the question?

Hint: $n\bar{r}^2$ is computed from all points before any question is asked.

Box D5: from regression to classification

Differentiate $L = \frac{1}{2}(y - F)^2$ once with respect to F to get the slope g, and again to get the curvature h. The best leaf value is $\gamma = -\frac{\sum g}{\sum h}$. Check that this gives the average residual. Now imagine a loss whose slope is $g = p - y$ and curvature $h = p(1 - p)$ (the log loss of classification). What would γ be?

Hint: this is the leaf value of the classification document; with $h = 1$ the denominator is just n.

7. The recipe and common confusions

Gradient boosting for regression in five lines.

0. Start with F_0 = the average of y.
1. Compute the residuals $r_i = y_i - F(x_i)$.
2. Choose the question with the largest score $\frac{(\sum_L r)^2}{n_L} + \frac{(\sum_R r)^2}{n_R}$.
3. Set each leaf to the average residual of its points.
4. Update $F \leftarrow F + \nu \cdot$ (leaf value) and go to 1.

"Why fit the residuals and not y again?" A tree fitted to y would just repeat the same staircase. The residuals are exactly what the current model does *not* yet know; fitting them adds new information.

"Why is it called *gradient* boosting?" Because the residual is the negative slope (gradient) of the loss with respect to the prediction (Box D1). Adding the fitted residual is a step of gradient descent, with the prediction values themselves as the thing being adjusted.

"Why a stump and why only a fraction ν?" A stump is a simple, weak model; many weak models added with small steps behave well. A big tree with a full step would fit the noise of the training points in one go.

"When do we stop?" The training error keeps falling (here from 50 to 0.867 after 8 rounds), but a model that follows every wiggle of the training points may predict new points badly. In practice the number of rounds is chosen on separate validation data.

"How is it different from AdaBoost?" AdaBoost passes the mistakes on as *weights* and votes; gradient boosting passes them on as *residuals* (more precisely, as slopes of the loss) and adds. With another loss (log loss) the same recipe does classification: see the classification document.

8. Check yourself

Fill in the blanks without looking, then check against the answers below.

1. The first guess F_0 is the _____ of the y values, here _____.

2. The residual of a point is $r =$ _____; a positive residual means the prediction was too _____.

3. A stump's leaf holds the _____ of the residuals of the points in the leaf.

4. Score of a question: $\frac{(\sum_L r)^2}{___} + \frac{(\sum_R r)^2}{___}$; the question with the _____ score is chosen.

5. Round 1 chooses $x \le$ _____ with $\gamma_L =$ _____ and $\gamma_R =$ _____.

6. With $\nu = 0.5$, the prediction of point 6 after round 1 is _____.

7. The sum of squared residuals is 50 at the start and _____ after round 1.

8. With $\nu = 1$ a stump reduces the sum of squared residuals by $n_L \bar{r}_L^2 + n_R \bar{r}_R^2 =$ _____ in round 1, leaving _____.

9. The residual is the _____ of the squared-error loss with respect to the prediction.

10. For a general loss the leaf value is $\gamma = -\sum g \,/\, \sum$ ____; for squared error $h =$ ____, so γ is the average residual.

Answers

1. average (mean); 5.5

2. $y - F$; low

3. average (mean)

4. n_L; n_R; largest

5. 4.5; -2.25; 2.25

6. 6.625

7. 19.625

8. 40.5; 9.5

9. negative slope (negative gradient)

10. h; 1

9. Solutions to the boxes

Box 1

$S(c) = \frac{1}{2}\sum(y_i - c)^2 \Rightarrow S'(c) = -\sum(y_i - c) = 0 \Rightarrow \sum y_i = Nc \Rightarrow c = \frac{1}{N}\sum y_i$, the average. ($S'' = N > 0$: a minimum.)

Your turn 1

$x \le 2.5$: yes-group points 1, 2 with residuals $-3.5, -2.5$: sum -6, $n_L = 2$; no-group points 3–8: sum $+6$, $n_R = 6$. Score $= \frac{36}{2} + \frac{36}{6} = 18 + 6 = 24$. Smaller than the best score 40.5 (question $x \le 4.5$): its two groups are less pure.

Your turn 2

Point 6 is in the no-leaf: $F_1(6) = 5.5 + 0.5(2.25) = 6.625$ with $\nu = 0.5$; with $\nu = 1$: $5.5 + 2.25 = 7.75$. The truth is 8, so the full step is closer in this one round; but after more rounds with $\nu = 0.5$ the prediction keeps improving (after round 3 it is 7.735), and the small steps protect us when one stump is misled by noise.

Your turn 3

Residuals after round 1 (table): points 1, 2, 3: -2.375, -1.375, -1.375, mean $-1.708 = \gamma_L$. Points 4–8: 0.625, -0.625, 1.375, 1.375, 2.375, mean $1.025 = \gamma_R$. Point 5: $F_1(5) = 6.625$, so $F_2(5) = 6.625 + 0.5(1.025) = 7.138$.

Box D1

$\frac{dL}{dF} = (y - F)\cdot(-1) = -(y - F) = -r$. The slope is $-r$: increasing F by a little changes L by $-r$ times that little. If $r > 0$ (prediction too low) the loss falls when F rises, so we should move F *up*, in the direction of the residual. The residual is the negative gradient.

Box D2

$\frac{d}{d\gamma}\frac{1}{2}\sum(r_i - \gamma)^2 = -\sum(r_i - \gamma) = 0 \Rightarrow \gamma = \frac{1}{n}\sum_{i\in\text{leaf}} r_i = \bar{r}$, the average residual of the leaf.

Box D3

For one leaf, $\sum(r_i - \bar{r})^2 = \sum r_i^2 - 2\bar{r}\sum r_i + n\bar{r}^2 = \sum r_i^2 - n\bar{r}^2$. After the split, the squared error is $\sum r_i^2 - n_L\bar{r}_L^2 - n_R\bar{r}_R^2$ (both leaves at their means). Before the split, with a single leaf at the overall mean $\bar{r} = 0$ (the residuals of the average add up to 0), it is $\sum r_i^2$. So the fall is $n_L\bar{r}_L^2 + n_R\bar{r}_R^2 = \frac{(\sum_L r)^2}{n_L} + \frac{(\sum_R r)^2}{n_R}$. Round 1: $4(-2.25)^2 + 4(2.25)^2 = 20.25 + 20.25 = 40.5$, and $50 - 40.5 = 9.5$ is the new squared error with a full step ($\nu = 1$), as computed by the program.

Box D4

In general (when the overall mean residual is not 0) the fall is $\frac{(\sum_L r)^2}{n_L} + \frac{(\sum_R r)^2}{n_R} - n\bar{r}^2$. The term $n\bar{r}^2$ is the same for every candidate question, so to choose the question we only need to **maximise** $\frac{(\sum_L r)^2}{n_L} + \frac{(\sum_R r)^2}{n_R}$. This is the score of Section 3.

Box D5

Differentiating, $g = \frac{d}{dF}\frac{1}{2}(y-F)^2 = -(y-F) = -r$ and $h = \frac{d}{dF}[-(y-F)] = 1$. Then $\gamma = -\frac{\sum g}{\sum h}$ gives $\gamma = \frac{\sum r}{n} = \bar{r}$: the average residual. For the log loss, $g = p - y = -(y-p)$ and $h = p(1-p)$, so $\gamma = \frac{\sum(y-p)}{\sum p(1-p)}$: the same recipe with a different curvature (the classification document derives and uses it).